

OIL INDUSTRY SAFETY DIRECTORATE
OISD-STD-105 (SYNTHETIC REFERENCE DOCUMENT)
HOT WORK ? PERMIT TO WORK SYSTEM

Third Revision ? 2018

NOTE: This is a synthetic document prepared for training and demonstration purposes, structured to reflect the format and requirements of Indian industrial safety standards. It is not a reproduction of any official government publication.

CHAPTER 1 ? SCOPE AND APPLICATION

- 1.1 This standard specifies minimum requirements for the Permit to Work (PTW) system applicable to all hot work operations conducted in oil and gas installations, refineries, petrochemical plants, and other hazardous area facilities in India.
- 1.2 Hot work, for the purpose of this standard, includes: welding, gas cutting, plasma cutting, grinding (that produces sparks), operation of internal combustion engines in hazardous areas, and any other work that may produce heat, sparks, or flame.
- 1.3 This standard shall be read in conjunction with OISD-STD-116 (Fire Protection Facilities), OISD-STD-144 (Liquefied Petroleum Gas), and the applicable provisions of the Factories Act, 1948 and the Petroleum Act, 1934.

CHAPTER 2 ? DEFINITIONS

- 2.1 Hot Work Area: Any area where hot work is to be performed.
- 2.2 Hazardous Area: An area where flammable or explosive atmospheres may exist.
- 2.3 Lower Explosive Limit (LEL): The minimum concentration of a flammable gas or vapour in air below which the mixture cannot be ignited.
- 2.4 Permit Issuing Authority (PIA): The person authorised to issue a hot work permit, typically the area supervisor or safety officer. The PIA must have completed hot work permit training and be assessed as competent.
- 2.5 Permit Receiver: The person performing or supervising the hot work.

CHAPTER 3 ? GENERAL REQUIREMENTS FOR HOT WORK PERMITS

- 3.1 No hot work shall be performed without a valid Hot Work Permit issued by the Permit Issuing Authority.
- 3.2 A Hot Work Permit shall be valid for one shift only (maximum 12 hours). If work extends beyond one shift, a fresh permit must be issued.
- 3.3 The Permit Issuing Authority shall conduct a physical inspection of the work area before issuing the permit.
- 3.4 The Hot Work Permit shall specify:
 - (a) Location of work (area, equipment tag, elevation)
 - (b) Nature of hot work to be performed
 - (c) Duration of permit validity
 - (d) Gas test results at time of permit issuance
 - (e) Name and signature of Permit Issuing Authority
 - (f) Name and signature of Permit Receiver
 - (g) Fire extinguisher location and standby arrangements
 - (h) Conditions under which hot work must be immediately stopped

CHAPTER 4 ? GAS TESTING REQUIREMENTS

4.1 GAS TEST BEFORE PERMIT ISSUANCE

- 4.1.1 A gas test shall be performed in the hot work area immediately before permit issuance. The gas test must be conducted by a competent person using a calibrated portable gas detector.
- 4.1.2 Hot work shall NOT be permitted if the combustible gas concentration in the work area exceeds 10% LEL.